MECHANICAL ENGINEERING PROJECT REPORT

PROJECT TITLE

DESIGN AND STATIC STRUCTURAL ANALYSIS OF A CONNECTING - ROD USING SOLIDWORKS AND ANSYS

By

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Software Used

• SolidWorks 2025

• ANSYS Workbench 2026 R1

ABSTRACT

The connecting rod is a critical component of an internal combustion engine that transfers the reciprocating motion of the piston into rotational motion of the crankshaft. Due to continuous exposure to high mechanical loads during engine operation, the structural performance and reliability of the connecting rod are essential for efficient and safe engine functioning.

This project focuses on the design and static structural analysis of an automotive connecting rod using **SOLIDWORKS** and **ANSYS**. A detailed three-dimensional CAD model of the connecting rod was developed in SOLIDWORKS, followed by preparation of engineering drawings and simulation setup for structural evaluation. The finite element analysis was performed in ANSYS to study the behavior of the component under static loading conditions by evaluating important parameters such as equivalent (von Mises) stress, elastic strain, and total deformation.

The analysis provides an understanding of the stress distribution and deformation characteristics of the connecting rod, helping to evaluate the structural suitability of the design. The project demonstrates the application of computer-aided design (CAD) and computer-aided engineering (CAE) techniques in the development and validation of mechanical components used in automotive applications.

The successful completion of this study highlights the importance of simulation-based design approaches in modern mechanical engineering for improving component reliability, performance, and design optimization.

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1. INTRODUCTION

A connecting rod is an essential component of an internal combustion (IC) engine. It connects the piston to the crankshaft and transfers the force produced during combustion into rotary motion. During engine operation, the connecting rod experiences high compressive and tensile forces. Therefore, it must have sufficient strength, durability, and fatigue resistance to ensure reliable engine performance.

Computer-Aided Design (CAD) and Finite Element Analysis (FEA) are widely used in modern engineering to design and evaluate mechanical components. **SolidWorks** is used to create accurate 3D models, while **ANSYS Workbench** is used to analyze the structural behaviour of components under different loading conditions. This process helps engineers improve design quality while reducing manufacturing cost and development time.

In this project, a three-dimensional model of a connecting rod was created in **SolidWorks** and imported into **ANSYS Workbench** for static structural analysis. Material properties, boundary conditions, and external loading were applied to evaluate the component's performance.

The analysis focused on evaluating **Equivalent (von Mises) Stress**, **Total Deformation**, and the **Factor of Safety**. The results were used to identify critical stress regions and verify that the connecting rod design can safely withstand the applied load.

OBJECTIVES

- To design a 3D model of a connecting rod using SolidWorks.
- To import the CAD model into ANSYS Workbench.
- To assign appropriate material properties.
- To generate a finite element mesh.
- To apply boundary conditions and external load.
- To perform static structural analysis.
- To evaluate Equivalent (von Mises) Stress.
- To determine Total Deformation.
- To calculate the Factor of Safety.

2. DESIGN METHODOLOGY IN SOLIDWORKS

2.1 Sketching: Base Profile (Eye Ends, Shank, Fillets)

The modelling process began by creating the **primary sketch** of the connecting rod. The **big end (crank end)** and **small end (piston pin end)** were created using circles with the specified diameters. These two ends were connected using straight lines to form the **shank** of the connecting rod. Appropriate dimensions and geometric relations were applied to fully define the sketch. Fillets were added to provide smooth transitions between the shank and the eye ends, improving the overall geometry and reducing stress concentration.

After creating the main body of the connecting rod, a **second sketch** was created on the required face of the model. This sketch defined the profile for removing material from the lower portion of the connecting rod. The second sketch was then used with the **Extruded Cut** feature to obtain the final shape of the component according to the design requirements.

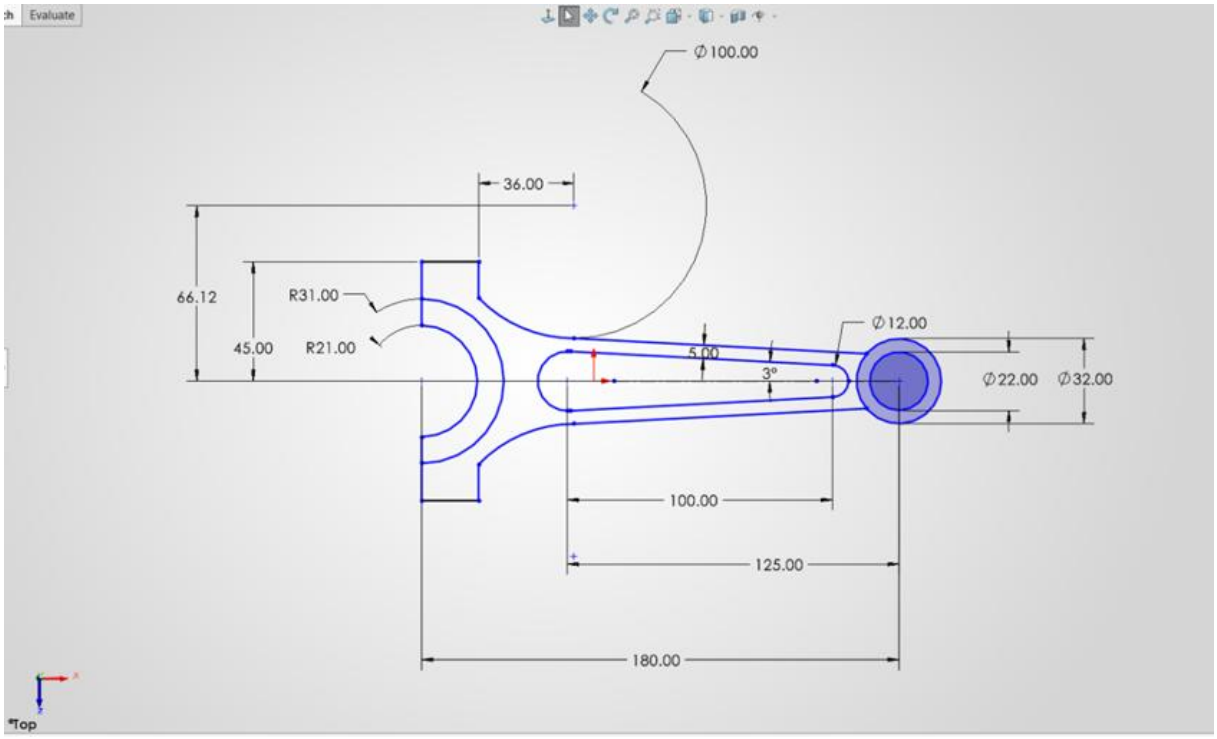


Figure 2.1: *Base profile sketch of the connecting rod.*

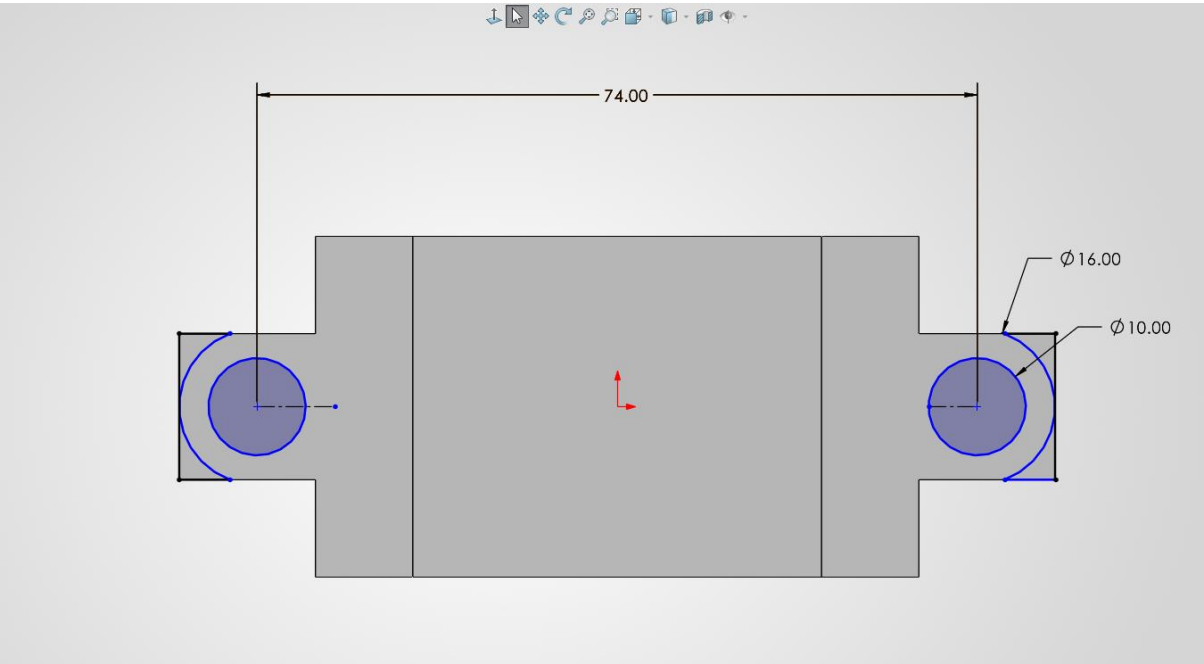


Figure 2.1(2): *Second sketch created for the Extruded Cut operation.*

2.2 Features: Extrude, Cut, Fillet, Chamfer

After completing the sketch, various SolidWorks features were used to create the final three-dimensional model.

- **Extrude Boss/Base** was used to convert the 2D sketch into a solid 3D model.
- **Extruded Cut** was used to remove material and create the required openings.
- **Fillet** was applied to sharp edges to improve the component's strength and reduce stress concentration.
- **Chamfer** was added to selected edges where required to improve the overall geometry and manufacturing feasibility.
- Additional modelling operations were performed to ensure that the final model matched the given dimensions.

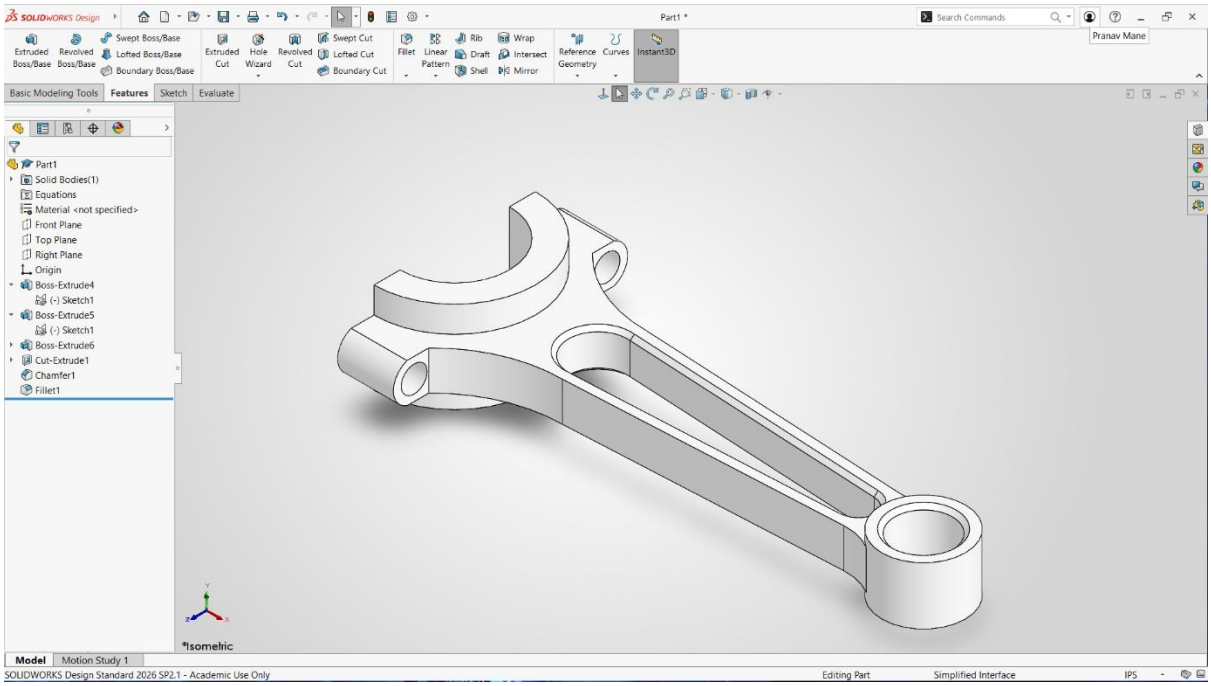


Figure 2.2: Final 3D connecting rod model developed using SolidWorks features.

2.3 Assembly Concept: Piston Pin End and Crank End

The connecting rod consists of two functional ends. The **small end** is connected to the **piston pin**, allowing the piston to transfer reciprocating motion to the connecting rod. The **big end** is connected to the **crankshaft journal**, which converts the reciprocating motion into rotary motion. Although this project focused on designing a single connecting rod component, the model was developed considering its position and function within the complete engine assembly.

2.4 Export as STEP/IGES for Simulation

After verifying the dimensions and geometry, the completed connecting rod model was saved in the native **SolidWorks (.SLDPRT)** format. To perform structural analysis in ANSYS Workbench, the model was exported as a **STEP (.STP)** file. The STEP format preserves the model geometry accurately and ensures compatibility between SolidWorks and ANSYS.

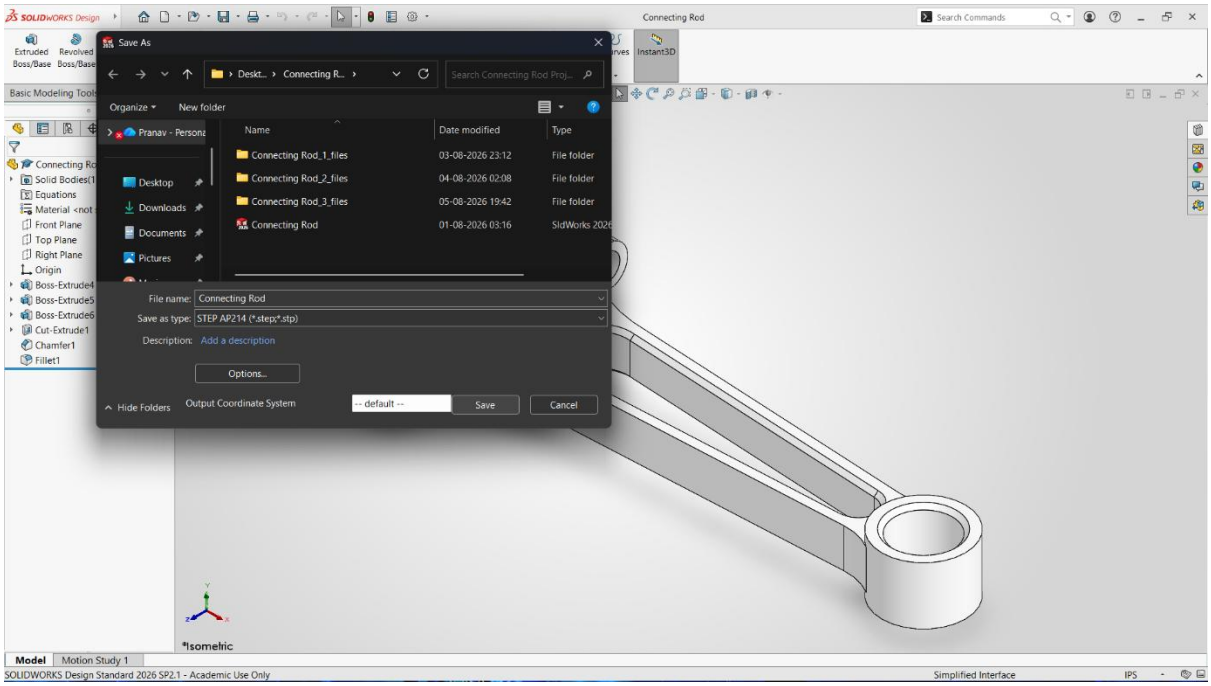


Figure 2.4: Exporting the CAD model in STEP format for ANSYS Workbench.

3. FINITE ELEMENT ANALYSIS IN ANSYS WORKBENCH

3.1 Geometry Import: From SolidWorks (STEP)

The completed connecting rod model was exported from SolidWorks in **STEP (.stp)** format and imported into **ANSYS Workbench**. The STEP format preserves the geometry accurately and ensures compatibility between CAD and simulation software. After importing, the geometry was verified before proceeding with the analysis.

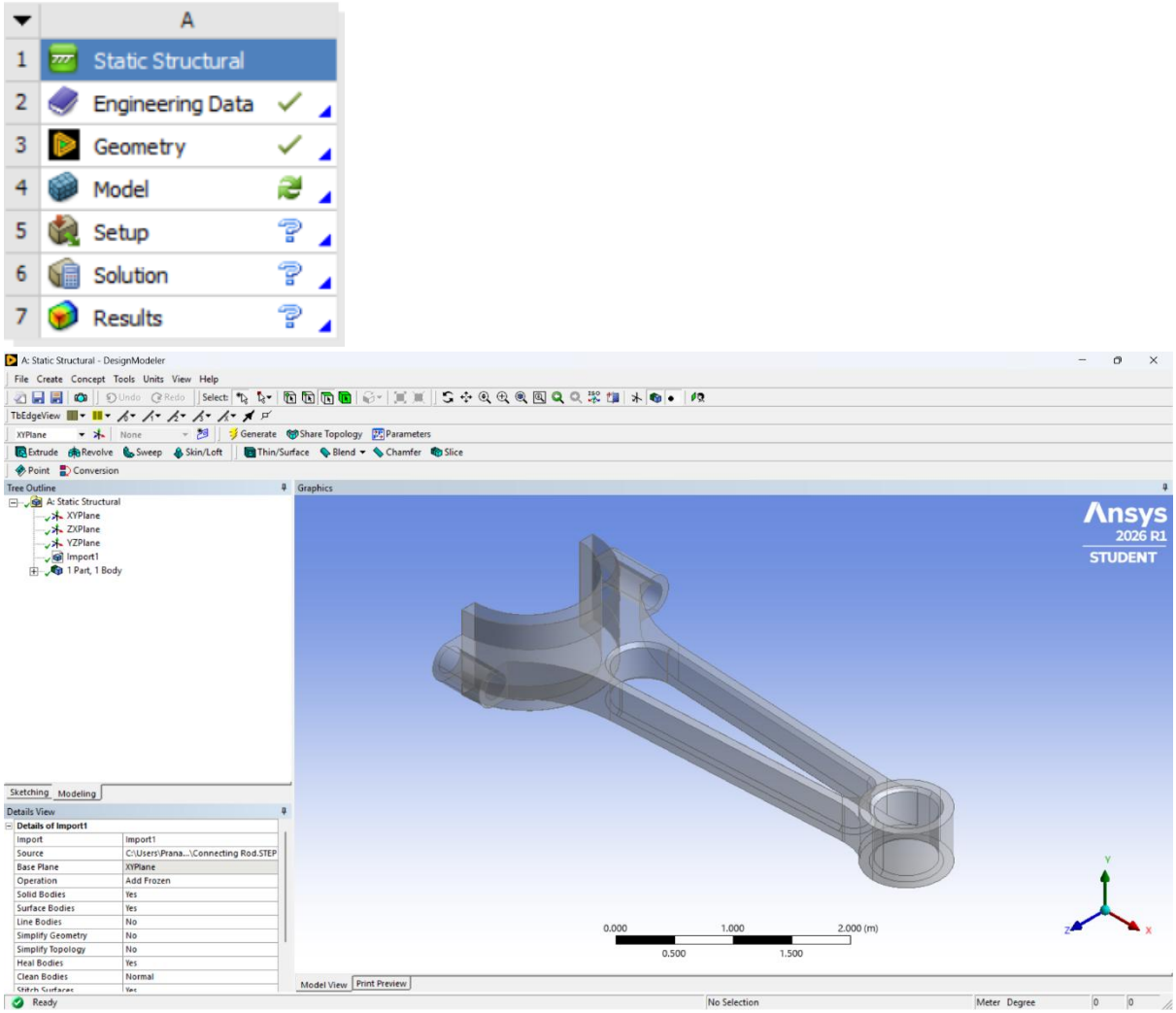


Figure 3.1: *Importing the connecting rod geometry into ANSYS Workbench.*

3.2 Material Assignment: Steel (E = 200 GPa, ν = 0.3)

The material assigned to the connecting rod was **Structural Steel**. The material properties used for the analysis were:

- Young's Modulus (E) = **200 GPa**
- Poisson's Ratio (ν) = **0.30**

These properties represent the elastic behaviour of steel during static loading conditions.

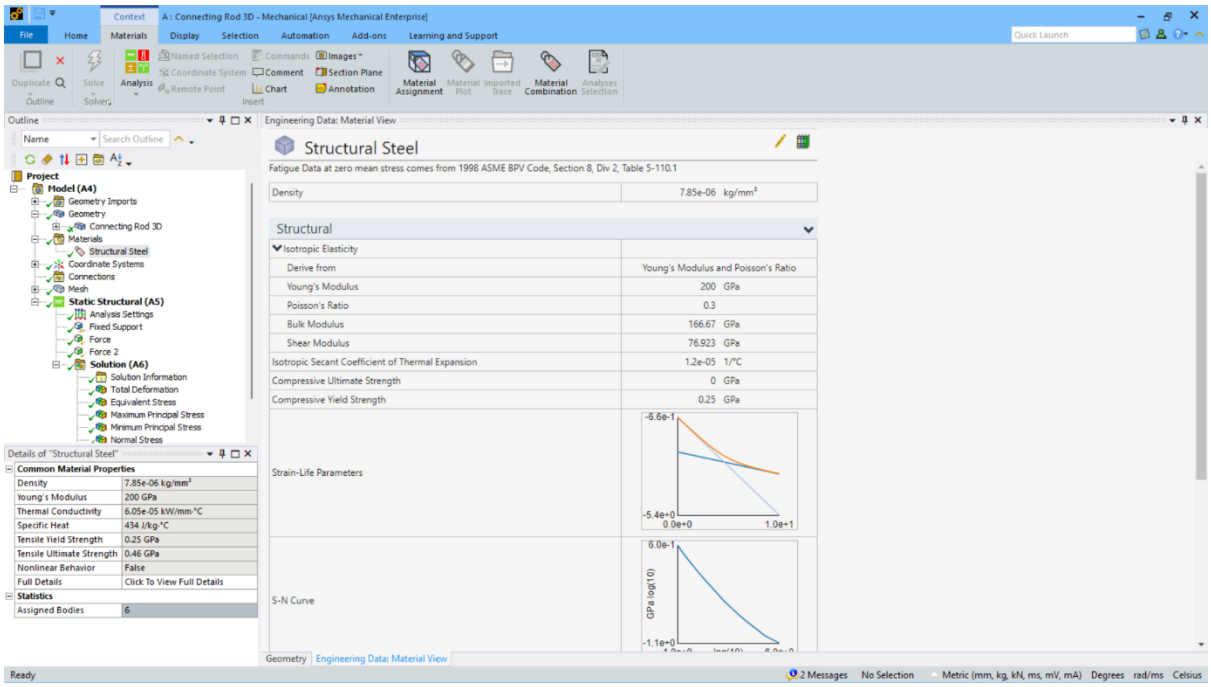


Figure 3.2: *Structural Steel assigned to the connecting rod.*

3.3 Meshing: Fine Mesh at Fillets and Load Transfer Regions

A finite element mesh was generated to divide the connecting rod into small elements for numerical analysis. A relatively finer mesh was used around the fillet regions and load transfer areas to improve the accuracy of stress prediction because these regions experience higher stress concentrations.

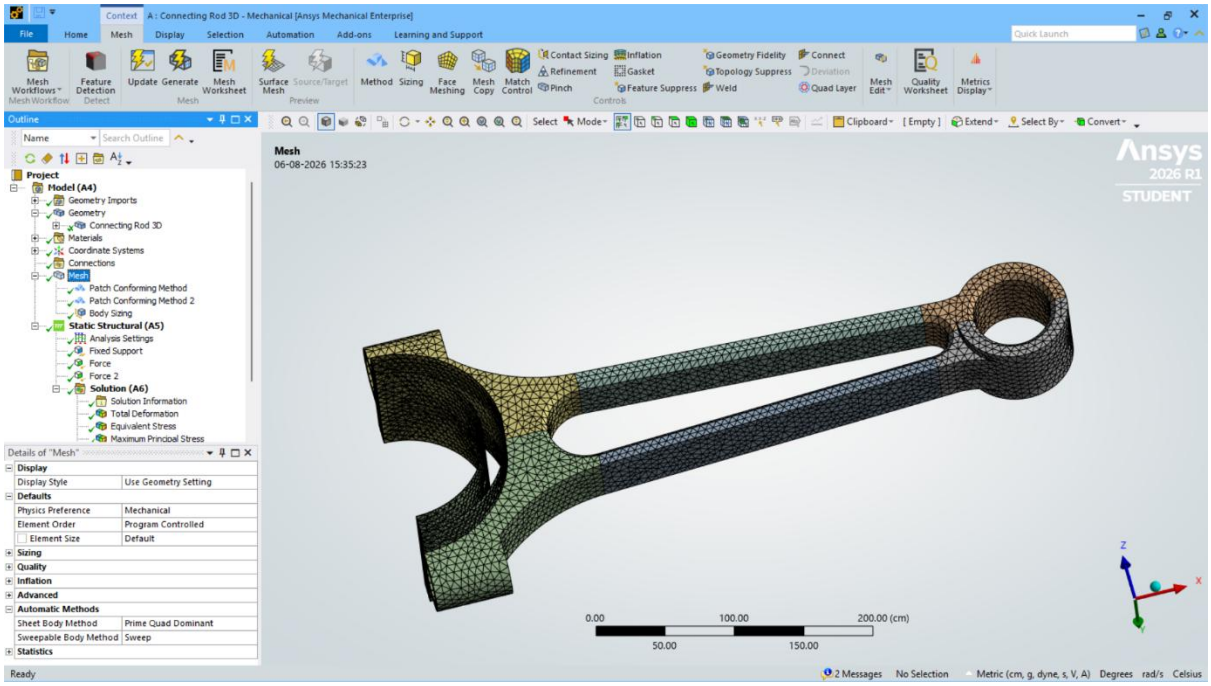


Figure 3.3: *Finite element mesh generated for the connecting rod.*

3.4 Boundary Conditions

Boundary conditions were applied to simulate the loading conditions of the connecting rod. The **big end (crankshaft side)** was constrained using a **Fixed Support**, while external forces were applied at the **small end (piston pin side)**. Two force components were considered to represent the loading condition:

- **X-direction: −5000 N**
- **Z-direction: −750 N**

The negative sign indicates that the forces were applied in the negative X and negative Z directions of the selected coordinate system. These boundary conditions were used to perform the static structural analysis.

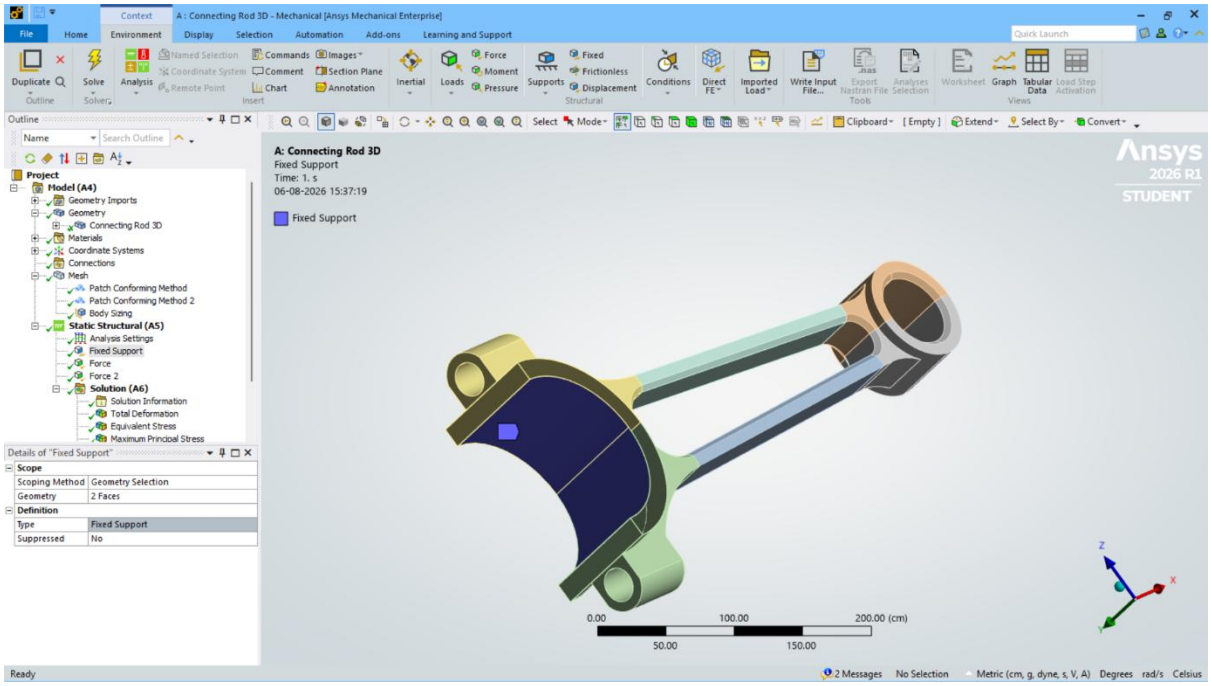


Figure 3.4(1): *Fixed support applied at the big end (crankshaft side) of the connecting rod.*

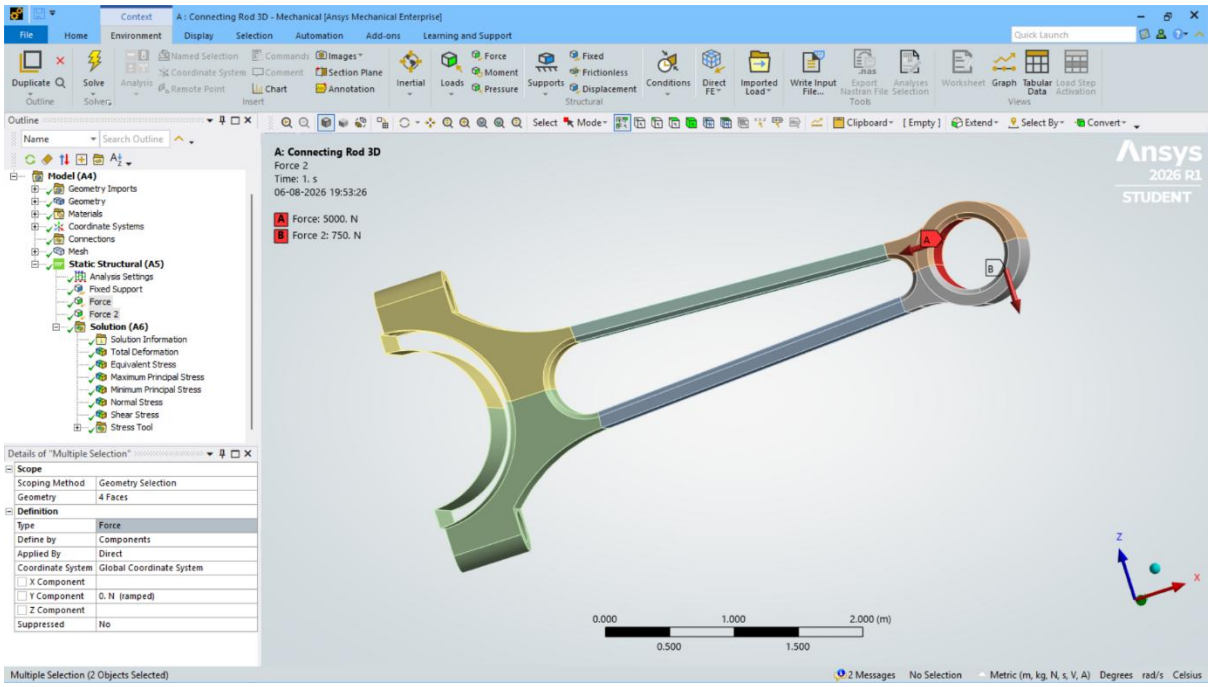


Figure 3.4(2): *External force applied at the small end (piston pin side) of the connecting rod.*

3.5 Analysis Type: Static Structural

A **Static Structural Analysis** was performed to evaluate the response of the connecting rod under the applied load. This type of analysis assumes that the applied load remains constant with time and is suitable for determining stress, deformation, and safety factor under static loading conditions.

3.6 Outputs

The following results were obtained from the static structural analysis:

- Total Deformation
- Equivalent (von Mises) Stress
- Safety Factor

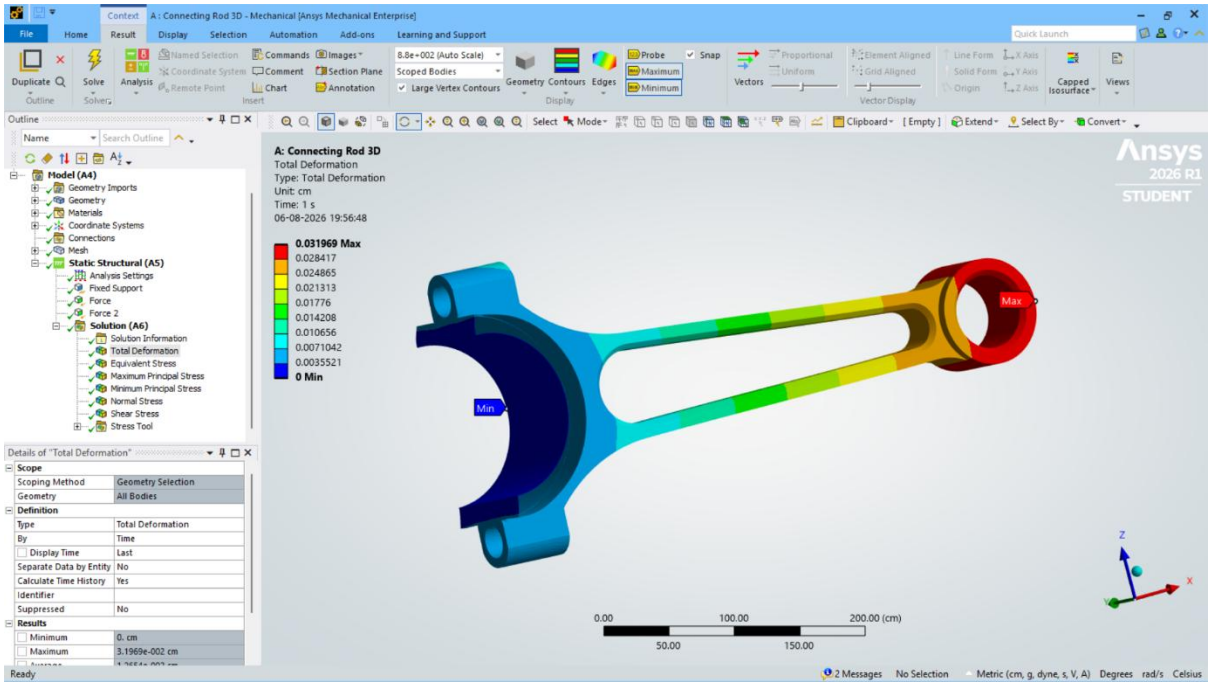


Figure 3.6(1): *Total deformation of the connecting rod.*

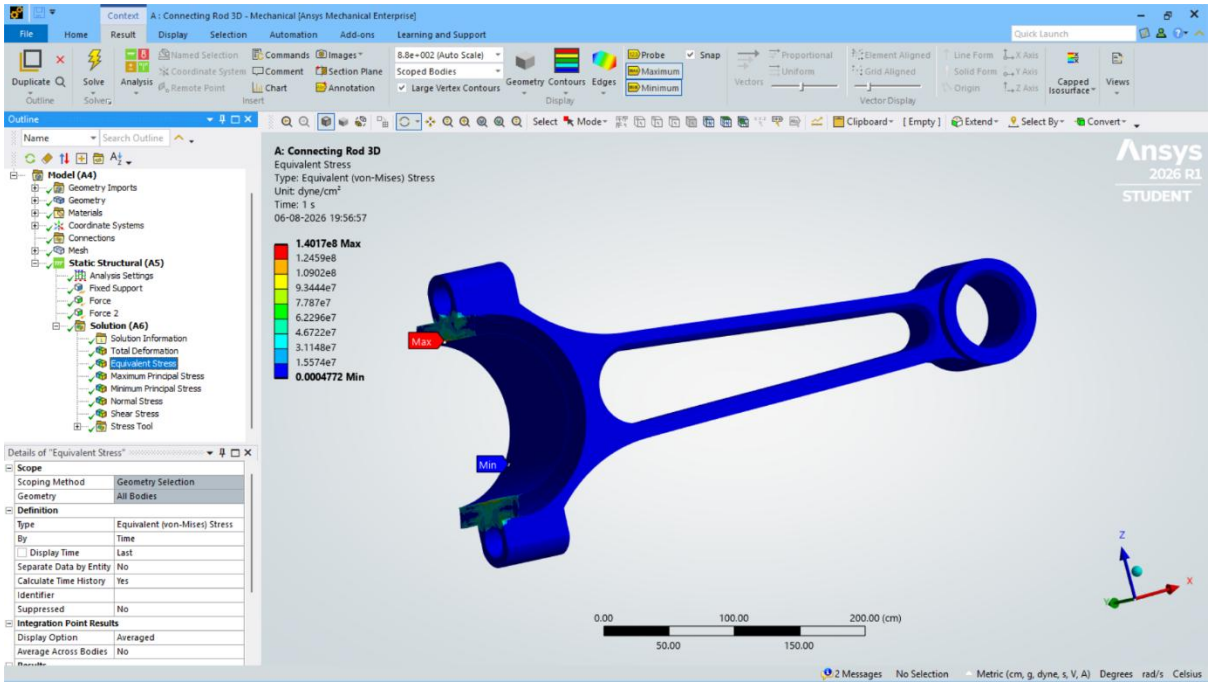


Figure 3.6(2): *Equivalent (von Mises) stress distribution in the connecting rod.*

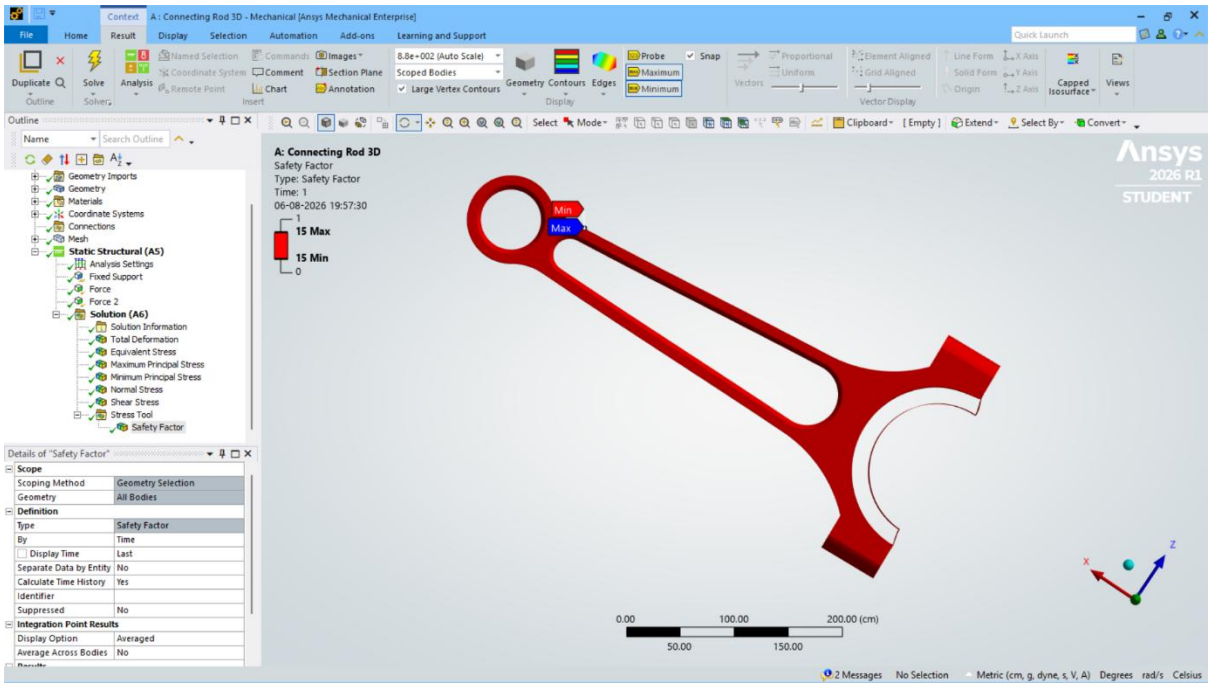


Figure 3.6(3): *Factor of Safety of the connecting rod.*

4. RESULTS AND DISCUSSION

After completing the static structural analysis of the connecting rod in ANSYS Workbench, the obtained results were evaluated to determine the deformation, stress distribution, and structural safety of the component. The analysis helps in identifying the critical regions where maximum stress occurs and verifies whether the connecting rod can safely withstand the applied loading conditions.

4.1 Total Deformation Analysis

The total deformation result represents the displacement of the connecting rod due to the applied external forces.

The ANSYS analysis shows that the **maximum total deformation is 0.3197 mm**, while the **minimum deformation is 0 mm** at the fixed support.

The maximum deformation occurs at the **small end (crank/piston pin end)** of the connecting rod because this end is subjected to the applied loading, whereas the big end is constrained by the fixed support.

Since the deformation is very small, it indicates that the connecting rod possesses sufficient stiffness and can withstand the applied load without experiencing excessive displacement.

Result

Parameter	Value
Maximum Total Deformation	0.3197 mm
Minimum Total Deformation	0 mm
Location of Maximum Deformation	Small End (Crank/Piston Pin End)

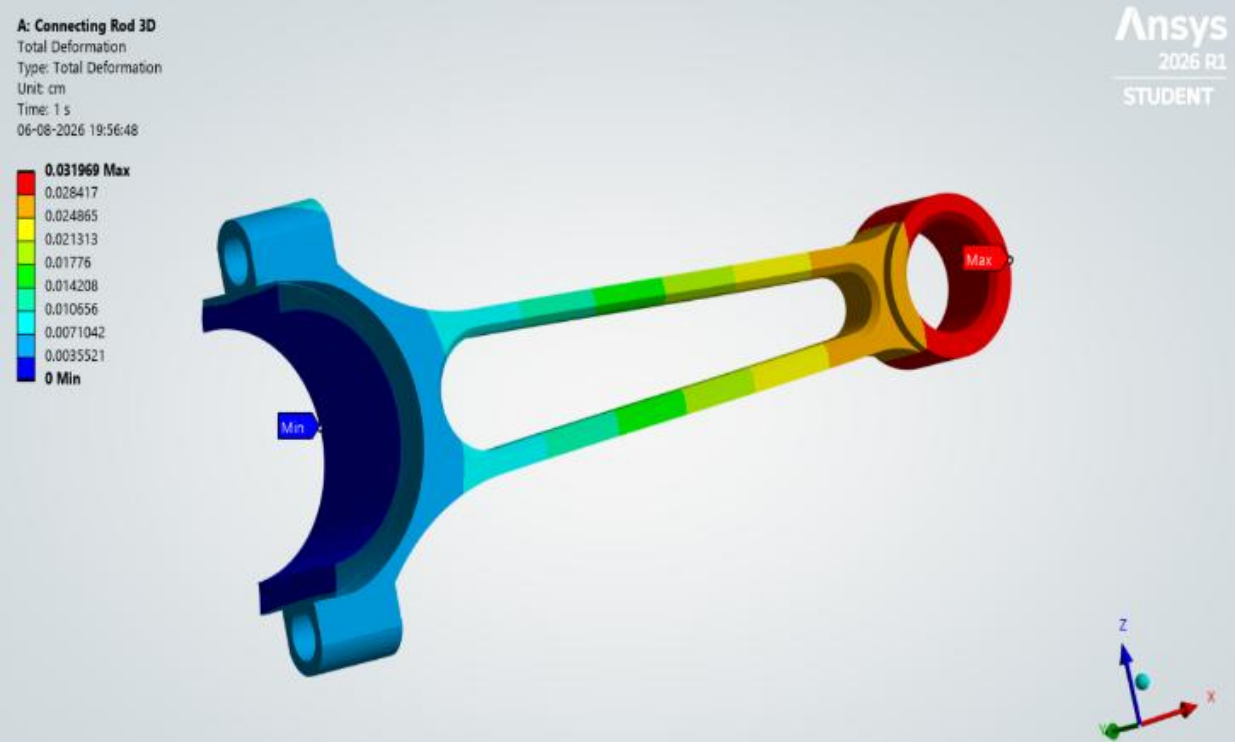


Figure 4.1: *Total deformation of the connecting rod under the applied loading condition.*

4.2 Equivalent (Von-Mises) Stress Analysis

The Equivalent (Von-Mises) Stress is used to predict whether the component will yield under the applied loading condition.

From the ANSYS simulation, the **maximum Equivalent (Von-Mises) stress is 14.02 MPa**, whereas the minimum stress is nearly zero.

The highest stress is observed around the **fillet region near the big end** of the connecting rod. This region experiences stress concentration because of the sudden change in geometry between the shank and the bearing portion.

The remaining areas of the connecting rod experience comparatively lower stresses, indicating that the load is distributed effectively throughout the component.

Result

Parameter	Value
Maximum Von-Mises Stress	14.02 MPa
Minimum Von-Mises Stress	Approximately 0 MPa
Critical Region	Fillet near the Big End

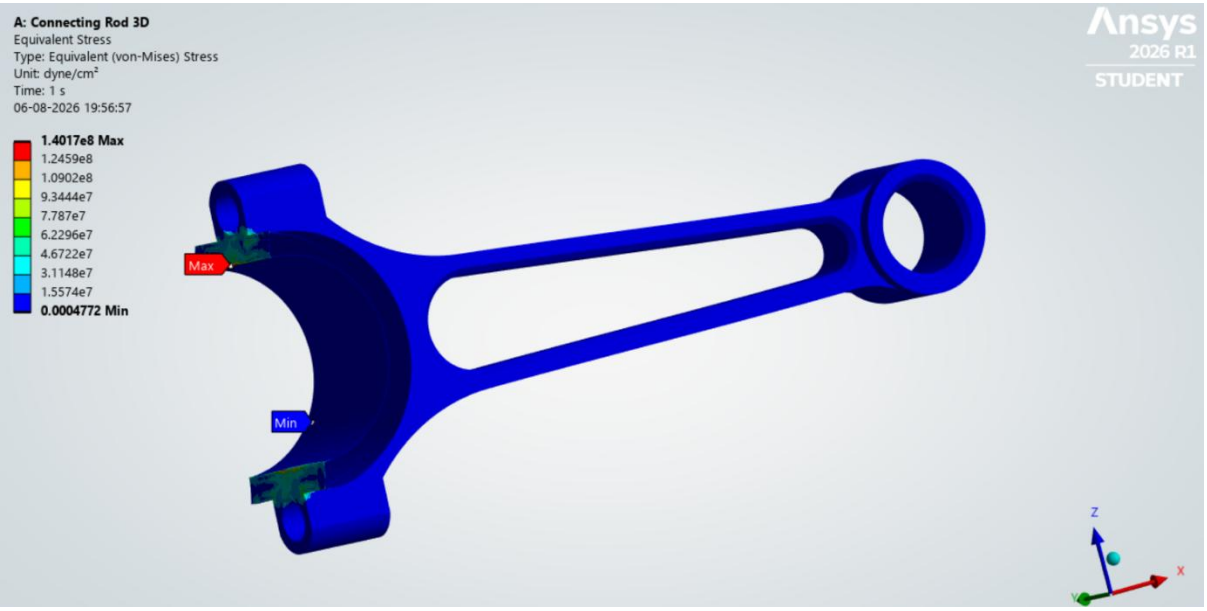


***Figure 4.2**: Equivalent (Von-Mises) stress distribution showing the highest stress concentration near the fillet region.*

4.3 Comparison with Material Yield Strength

The selected material for the connecting rod is **Structural Steel**.

The yield strength of Structural Steel is approximately **250 MPa**.

The maximum stress obtained from the ANSYS analysis is only **14.02 MPa**, which is significantly lower than the material yield strength.

Stress Comparison

Property	Value
Material	Structural Steel
Yield Strength	250 MPa
Maximum Von-Mises Stress	14.02 MPa
Difference	235.98 MPa

Since

14.02 MPa < 250 MPa

the connecting rod operates safely within the elastic region and no permanent deformation or yielding is expected under the applied loading condition.

4.4 Identification of Critical Stress Regions

During the finite element analysis, stress concentration was observed at specific locations due to changes in geometry.

The critical regions identified are:

- Fillet region near the big end.
- Bearing seating area around the big end pin hole.
- Transition section between the shank and the big end.

These regions experience higher stresses because stress concentration naturally occurs where there is a sudden variation in cross-section. Therefore, these areas require careful attention during design and manufacturing.

4.5 Factor of Safety

The Factor of Safety (FoS) obtained from the ANSYS analysis is **15**.

A factor of safety greater than 1 indicates that the component is safe under the applied loading condition.

Since the obtained FoS is **15**, the connecting rod has a very high safety margin and is capable of carrying the applied load without failure.

Result

Parameter	Value
Factor of Safety	15
Design Status	Safe

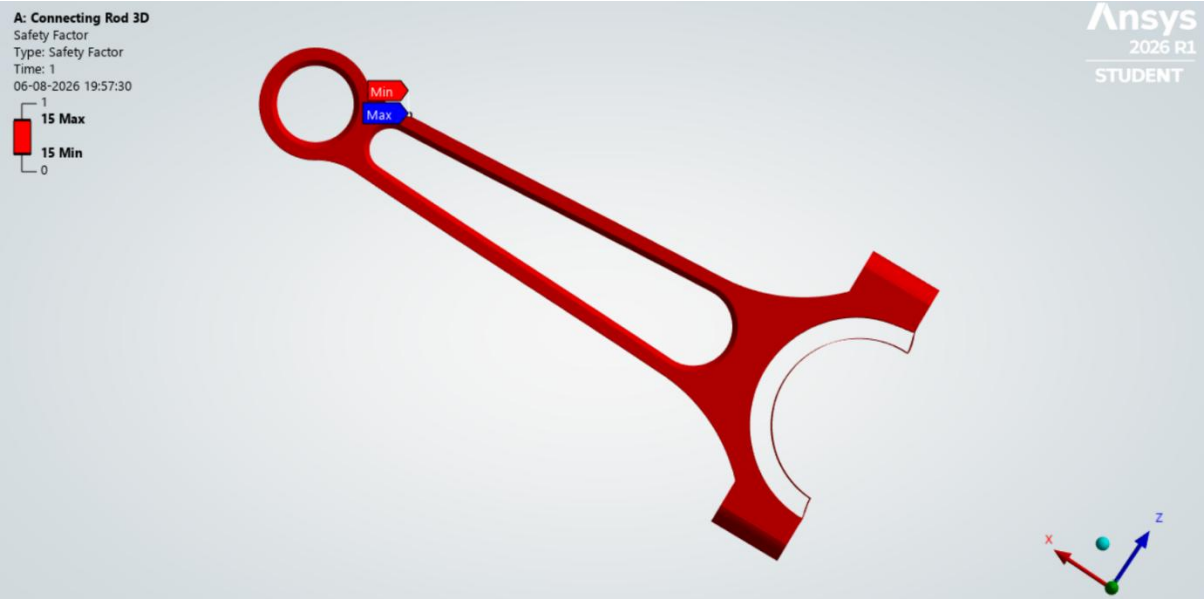


***Figure 4.5**: Factor of Safety distribution of the connecting rod.*

4.6 Discussion

The finite element analysis confirms that the connecting rod performs safely under the given loading conditions.

The deformation obtained is only **0.3197 mm**, indicating that the component has good stiffness and experiences very little displacement during operation.

The maximum Equivalent (Von-Mises) stress of **14.02 MPa** is much lower than the material yield strength of **250 MPa**, which confirms that the connecting rod remains within the elastic limit and will not undergo permanent deformation.

Although the highest stress is observed at the fillet region near the big end, the stress level is well below the allowable limit. The Factor of Safety of **15** further confirms that the connecting rod design is structurally safe and suitable for the specified loading conditions.

Overall, the ANSYS simulation verifies that the designed connecting rod has adequate strength, stiffness, and reliability for static loading.

4.7 Final Results Summary

Parameter	**Result**
Maximum Total Deformation	0.3197 mm
Maximum Von-Mises Stress	14.02 MPa
Material Yield Strength	250 MPa
Critical Stress Region	Fillet near Big End
Maximum Deformation Location	Small End (Crank/Piston Pin End)
Factor of Safety	15
Design Status	Safe

5. CONCLUSION

This project successfully accomplished the design and static structural analysis of an automotive connecting rod using **SOLIDWORKS** for three-dimensional CAD modeling and **ANSYS** for finite element analysis. The project followed a systematic engineering approach, beginning with the creation of an accurate CAD model, preparation of detailed engineering drawings, material assignment, mesh generation, application of static loading and boundary conditions, and evaluation of the simulation results.

The static structural analysis provided a comprehensive understanding of the mechanical behavior of the connecting rod by evaluating the distribution of equivalent (von Mises) stress, elastic strain, and total deformation under the specified loading conditions. The obtained results demonstrated the structural response of the component and verified its performance within the defined operating conditions. The analysis also highlighted the importance of proper geometry, material properties, and loading conditions in ensuring the structural integrity and reliability of mechanical components.

This project demonstrates the practical application of computer-aided design (CAD) and computer-aided engineering (CAE) tools in the design, analysis, and validation of automotive components. It also illustrates the integration of engineering design principles, material selection, finite element methods, and technical documentation into a structured product development process.

The successful completion of this project has provided valuable experience in engineering design methodology, structural simulation, result interpretation, and technical reporting. The knowledge gained through this work establishes a strong foundation for undertaking more advanced mechanical design, product development, and automotive engineering projects in both academic and industrial environments.

6. REFERENCES

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